Checklist for My Portfolio for ITRI626

SU2 NEURAL NETWORKS - INTRODUCTION

At least:

- Your assignment submitted on eFundi
  Simple FF networks explained with creative scenarios

Optional:

- Other Jupyter notebooks that you tried or learned from

SU3 LEARNING MODELS – BASIC APPLICATIONS

Chapters 4 and 5 in Trask

At least:

- [original file] Chapter 3 - Forward Propagation - Intro to Neural Prediction.ipynb
- [with my comments] Chapter 3 - Forward Propagation - Intro to Neural Prediction.ipynb
- [original file] Chapter 4 - Gradient Descent - Into to Neural Learning.ipynb
- [with my comments] Chapter 4 - Gradient Descent - Into to Neural Learning.ipynb

Optional:

- Other Jupyter notebooks that you experimented with or learned from
- Chapter 5 - Generalizing Gradient Descent - Learning Multiple Weights at a Time.ipynb
- Chapter 6 - Intro to Backpropagation - Building your first DEEP Neural Network.ipynb
- Chapter 8 - Intro to Regularization - Learning Signal and Ignoring Noise.ipynb
- Chapter 9 - Intro to Activation Functions - Learning to Model Probabilities.ipynb

SU4 - CONVOLUTIONAL NEURAL NETWORKS

At least:

- Group project report
- Group project presentation slides
- All the experiments referred to in the Group project report
-- scripts / programs / notebooks / results / comments
- Dataset(s) in different stages of cleanup

Optional:

- More experiments related to the Group project (not in report)
- Other Jupyter notebooks that you experimented with or learned from
- Trask Chapter 10 - Intro to Convolutional Neural Networks - Learning Edges and Corners.ipynb
- Trask Chapter 11 - Intro to Word Embeddings - Neural Networks which Understand Language.ipynb

SU5, SU6, SU7 - TRANSFER LEARNING - LARGE LANGUAGE MODELS - AGENTS

At least:

- A document outlining your learning process
  - E.g. starting from the "Starting points" in the "Update for Study Units 5,6,7.pdf"
    - each model your downloaded / tried and what you learned
- Your system which uses a local LLM for agentic workflow(s) (Suggested platform: unsloth)
- Document your system completely, including
  - design and development
  - intro video and usage instructions

Summary of learning outcomes for ITRI626:

SU1 - MACHINE LEARNING
- Describe the concept of learning and learning approaches
- Describe the difference between parametric and non-parametric learning
- Describe the difference between machine learning paradigms in relation to how they treat data and their application spaces.
- Design and analyse machine learning algorithms

SU2 - NEURAL NETWORKS - INTRODUCTION
- Understand the basic mathematical model underlying the operation of a single neuron in a neural network
- Be able to describe
    - o Single-input, single-output network
    - o Multiple input, single output network
    - o Single input, multiple output network
    - o Multiple input, multiple output network

SU3 - LEARNING MODELS – BASIC APPLICATIONS
- Describe the black box model of machine learning algorithms
- Discuss the training, validation and testing functions of data
- Discuss the gradient descent algorithm for training models
- Discuss different neuron activation functions and their purposes
- Understand and be able to create regression models to solve problems using
    - o A linear regression model using Scikit-Learn
    - o Create a neural network to solve similar problems
- Understand and be able to create classification models to solve problems using
    - o Support vector classifier model using Scikit-Learn
    - o Create a neural network to solve similar problems
- Describe the full structure of a basic deep neural network

SU4 - CONVOLUTIONAL NEURAL NETWORKS
- Discuss the application areas of convolutional neural networks
- Discuss the function of a convolutional layer
- Discuss the function of a pooling layer

SU5, SU6, SU7 - TRANSFER LEARNING - LARGE LANGUAGE MODELS - AGENTS
- Be able to recognise the difference between using labelled and unlabelled data
- Understand the nature of data and how we can apply different techniques to unlabelled data than what we can to labelled data.
- Be able to perform clustering operations on unlabelled data
- Understand and be able to use tools that use semi-supervised learning
- Understand and be able to use transfer learning to apply a model that has been trained in one domain to a different domain.
- Understand the basic building blocks and operation of Large Language Models
- Be able to use Large Language Models effectively as they exist already
- Be able to fine-tune a Large Language Model to a specific domain